



CLÉMENT

Helsen

Senior Data Scientist & ML Engineer

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TECHNICAL STACK

ML & Vision

CoTracker3	ResNet	Deep Learning
Computer Vision		

Data & Infra

MariaDB	Spark	Docker	CI/CD
500 TB+			

Web & Viz

Django	Bokeh	Panel
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Languages

Python	C++	R	SQL
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Physics & Instruments

Geant4	Bayesian stats
Microscopy control	SLIMS

EDUCATION

- Ph.D. Particle Physics**
Univ. Paris XI & CEA Saclay
2006 – 2009
- M.Sc. Fundamental Physics**
University of Paris XI
2003 – 2006

HONORS

- 2025 Breakthrough Prize — Fundamental Physics
- 2013 Higgs boson discovery (co-author, ATLAS)

TEACHING & OUTREACH

- Supervision:** 5 PhD students & 2 postdocs (CERN, EPFL).
- Outreach:** CERN visitor guide; science talks in schools.

LANGUAGES

FR Native EN Fluent DE Intermediate

INTERESTS

Trail running & ultra-distance events, permaculture gardening.

PROFILE

Ph.D. physicist turned senior data scientist & ML engineer with **15+ years** shipping end-to-end data products across High Energy Physics, Life Sciences, and Sports Analytics. Deployed production computer-vision systems (CoTracker3, ResNet), managed **500 TB+** data pipelines, and built interactive analytics platforms used daily by 10+ researchers. Strong across the full ML lifecycle — data modelling and feature engineering through training, evaluation, and production deployment. Solid background in Bayesian and frequentist statistical modelling; author of open-source tooling adopted by multiple international collaborations.

EXPERIENCE

- **Independent Contractor** 2026 – present

Sports Performance & Nutrition · Self-employed

Sports Performance Platform (Django): Full-stack app integrating HRV, continuous glucose monitoring, training logs, nutrition tracking, and a custom race pacing model — pilot with ~10 athletes, targeting ultra-distance events.
- **Scientific Data Manager & R&D Physicist** 2023 – present

EPFL · Lausanne, Switzerland

Computer-Vision System (CoTracker3 fine-tuned): end-to-end pipeline for live biological imaging — sub-second inference, from acquisition through tracking to interactive dashboard. In production for 12+ months.

Laser Ablation Control: Full Bokeh dashboard — region selection, laser parameterisation, real-time acquisition control — replacing manual workflows entirely.

Drug Screening Pipeline: automated phenotyping platform (zebrafish) — SLIMS ↔ VAST ↔ Leica integration with 3 ResNet models (orientation, QC, somite counting). **Reduced assay turnaround from days/weeks to minutes;** now one of the lab's flagship workflows.

Adaptive Acquisition (Nikon): two-stage pipeline — cell detection at 10× then optimised imaging path at 40× — sharply increases usable cells per imaging session.

Data Platform (500 TB+): built from scratch a MariaDB single-cell image database (300 k+ entries) and a raw-data catalog with metadata framework; currently leading lab-wide tape migration.

Researcher Tooling: Bokeh & Panel analytical dashboards used daily by 10+ scientists across the lab; full publication list below.
- **Research Physicist & Software Coordinator** 2013 – 2022

CERN · Geneva, Switzerland · FCC Study & ATLAS Experiment

Key4HEP (co-founder): built an open-source software platform from scratch, now the de-facto standard across multiple major detector projects. Coordinated a 15-researcher international team.

Tech Leadership: ran FCC computing (6 yr) and ATLAS analysis (3 yr) — 50+ researchers across continents, 4 major spec releases, long-term architectural roadmap.

FCC-ee Flavour Physics: led the prospect studies for τ-leptonic decays of B⁺ and B_c⁺ at the future FCC-ee collider — first-author publications in JHEP (2021) and EPJC (2024); inputs to the FCC Feasibility Study.

Energy-Frontier Physics: co-authored a comprehensive review of heavy-resonance searches at future hadron colliders (EPJC 2019), benchmarking BSM reach across HL-LHC and FCC-hh scenarios.

Detector & Computing R&D: hadronic shower characterisation for future colliders (JINST 2016); offline-computing requirements and software architecture for FCC-ee (EPJ Plus 2021).

Public speaking: ~30 invited talks at international scientific conferences (ICHEP, EPS, SUSY, Higgs Couplings, CHEF, La Thuile).
- **Research Associate** 2009 – 2013

IFAE · Barcelona, Spain

ATLAS Physics Measurements: co-authored multiple measurements at √s = 8 TeV — top-quark pair charge asymmetry (EPJC 2016), vector-like quark searches and four-top final states (JHEP 2015) — driving the statistical interpretation in the IFAE group.

C++ Statistical Framework: open-source frequentist hypothesis-testing library — adopted by 20+ teams, used in 15+ peer-reviewed studies.

pyFBU: first Python library for multi-channel Bayesian unfolding — adopted by 5 teams, cited in 6+ studies; pioneered PyMC use in the field.

Detector Calibration: built calorimeter calibration pipelines across 10 000+ channels of the ATLAS detector; underpinned the energy-scale consistency of the full Run-1 dataset.

SELECTED PROJECTS

Smart Microscopy

CoTracker3 real-time biological tracking in production; pre-print submitted.

Drug Screen Pipeline

3 ResNet models + SLIMS + VAST/Leica mapping for zebrafish phenotyping.

SELECTED PUBLICATIONS

LIFE SCIENCES · EPFL · 2023–2026

2026

Real-Time Multi-Position and Multi-ROI Tracking with LiLiTTool for Smart Light-Sheet Microscopy in Growing Samples

EPFL preprint · 1st author

doi:10.64898/2026.04.03.715281

2023

Clock-driven waves of Tbx6 expression prefigure somite boundaries

bioRxiv preprint

doi:10.1101/2023.11.09.566373

2023

Evaluation of genetic demultiplexing of single-cell sequencing data from model species

Life Science Alliance

doi:10.26508/Lsa.202301979

HIGH ENERGY PHYSICS · SELECTED

2024

Prospects for B_c^+ and $B^+ \rightarrow \tau^+ \nu_\tau$ at FCC-ee

Eur. Phys. J. C

doi:10.1140/epjc/s10052-024-12418-0

2021

Prospects for $B^+ \rightarrow \tau^+ \nu_\tau$ at FCC-ee

JHEP 12 (2021) 133

doi:10.1007/JHEP12(2021)133

2021

Offline computing resources for FCC-ee and related challenges

Eur. Phys. J. Plus

doi:10.1140/epjp/s13360-021-02189-y

2019

Heavy resonances at energy-frontier hadron colliders

Eur. Phys. J. C

doi:10.1140/epjc/s10052-019-7062-3

2016

Containment and resolution of hadronic showers at the FCC

JINST 11, P09012

doi:10.1088/1748-0221/11/09/P09012

2016

Measurement of the charge asymmetry in top-quark pair production in the lepton-plus-jets final state at $\sqrt{s} = 8$ TeV with the ATLAS detector

Eur. Phys. J. C · ATLAS Collaboration

doi:10.1140/epjc/s10052-016-3910-6

2015

Search for production of vector-like quark pairs and of four top quarks in the lepton-plus-jets final state at $\sqrt{s} = 8$ TeV with the ATLAS detector

JHEP 08 (2015) 105 · ATLAS Collaboration

doi:10.1007/JHEP08(2015)105